DiabAI

Diabetes Prediction System

Healthcare ML Application

Audit Release Summary (ARS) Report

Software Quality Assurance (SQA) Documentation

Document ID	ARS-DiabAI-v1.0
Version	1.0-Quality-Audit
Status	**Released - 29 March 2026**
Testing Type	Black-Box / Manual
Tools Used	Jira, Postman, Chrome DevTools, Python
Jira Epic Status	DS-1: DiabAI: SQA Audit **(Done)**
Bugs Documented	2 (DS-25), (DS-26)
Open Bugs	**Zero**
Total Test Cases	18/18 **(Passed)**
Total Stories	5 (DS-2, DS-8, DS-14, DS-17, DS-21)

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1. Release Audit Summary

The **DiabAI v1.0-Quality-Audit** release has successfully completed its full Software Quality Assurance audit cycle and is hereby formally declared release-ready.

1.1 Test Execution

- All 18 test cases across 5 User Stories were executed and tracked in Jira Sprint DS Sprint 1.
- All test cases were verified as **Done**.

1.2 Bug Tracking

- Two high-priority bugs were discovered during testing.
- Both were prioritised, resolved, and confirmed via regression testing.
- At the time of this report, there are **zero open defects**.

1.3 Release Management

- Jira release version `v1.0-Quality-Audit` was tagged and formally released on 29 March 2026.
- Jira Epic `DS-1` is in a **Done** state.
- No other bugs or open issues exist in the project.

1.4 Conditional PASS

- TC-15 (CI/CD Deployment Verification) recorded a **Conditional PASS**.
- **Cause:** Infrastructure-level constraint (Railway.app trial credit expiry), explicitly not a code defect.
- Application confirmed live and functional on **7 February 2026**. Release formally tagged **29 March 2026**.
- Deployment logs confirm **100% GitHub build success**.

> **Audit Verdict:** The DiabAI system is **APPROVED FOR RELEASE** under the `v1.0-Quality-Audit` tag. All quality gates are satisfied. The audit was conducted in full accordance with the `STP-DiabAI-v1.0` and `TCS-DiabAI-v1.0`.

2. Quality Gate Table

The following table documents the formal quality gate evaluation for the `v1.0-Quality-Audit` release. Each criterion was verified against live Jira data retrieved on **29 March 2026**.

Criteria	Status	Comments / Jira Evidence
All 18 Test Cases Executed	**PASS**	All 18 sub-tasks under DS-2, DS-8, DS-14, DS-17, DS-21 transitioned to `Done` in DS Sprint 1. Verified via JQL query `issuetype = Subtask AND project = DS`.
All 5 User Stories Closed	**PASS**	Stories DS-2, DS-8, DS-14, DS-17, DS-21 confirmed `Done`. Epic DS-1 also in `Done` state as fetched from Jira.
Zero High-Priority Open Defects	**PASS**	Full bug search `(issuetype = Bug AND project = DS)` returned exactly 2 bugs—DS-25 and DS-26. Both are `High` priority and both are `Done`. Zero open bugs remain.
Version Tagged in Jira Release	**RELEASED**	`v1.0-Quality-Audit` confirmed `released: true` with `releaseDate: 2026-03-29` in the Jira Release Management dashboard. Both DS-25 and DS-26 are tagged under this Fix Version.
High-Priority Bugs Closed Before Release	**PASS**	DS-25 transitioned to `Done` on 28 Mar 2026 (changelog entry #10400). DS-26 transitioned to `Done` on 28 Mar 2026 (changelog entry #10399). Both resolved prior to the 29 Mar 2026 release date.
Post-Fix Regression Completed	**PASS**	Full suite of 18 sub-tasks was re-executed after resolution of DS-25 and DS-26 per STP Section 4.4 Regression Policy. All previously failing TCs (TC-02, TC-03, TC-05, TC-07) passed on re-run.
Data Integrity: Test Data Isolated from Training Data	**PASS**	Black-box testing methodology was employed throughout. Testers submitted clinical parameters via the web UI without access to `model.pkl` internals, training weights, or the Pima Indians dataset used for model training. Test inputs (e.g. Glucose=148, BMI=33.6) are empirical patient profiles, not derived from the training set. No data leakage or contamination detected.

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Criteria	Status	Comments / Jira Evidence
Production Deployment Verified	**PASS (Cond.)**	Application confirmed live and functional at `web-production-efc90.up.railway.app` on 7 Feb 2026 with 100% GitHub CI/CD build success. Currently offline due to Railway trial credit expiry—classified as an environment-level limitation, *not a code defect*. See Section 5 (Residual Risk).
API Security: CORS / XSRF Protection Active	**PASS**	External Postman requests to `/predict` were rejected by CORS protection. No stack traces or internal paths exposed. Verified via TC-16 (DS-22).
No Unresolved Security Vulnerabilities	**PASS**	SECURITY.md audit conducted. System provides graceful error messages. No `venv` path or module structure information is leaked. Verified per STP Section 2.3.

3. Defect Summary Table

During the audit cycle, two high-priority defects (**DS-25** and **DS-26**) were identified and documented. Both followed a rigorous SQA lifecycle within the Jira DS project: discovered during test execution, filed with supporting screenshot evidence, and prioritized for the `v1.0-Quality-Audit` fix version. Following developer intervention, both defects underwent formal re-testing and regression confirmation. According to Jira transition logs, these issues were successfully resolved and moved to **DONE** status on **28 March 2026**, ensuring a zero-defect state prior to the final release on **29 March 2026.**

DS-25: Logical Validation Gap (High Priority)

- **Issue:** The system accepted an input of '0' for Glucose, BMI, and Age, which are medically impossible values. This resulted in skewed "Negative" predictions rather than validation errors.
- **Root Cause:** The Streamlit frontend sliders lacked a lower-bound constraint (*min_value*) for medically critical diagnostic fields.
- **Resolution:** Updated the `Web_Frontend` logic to enforce physiologically valid minimums (e.g., Glucose > 50, BMI > 10, Age ≥ 1).
- **Verification:** Verified via Boundary Value Analysis (BVA). The system now correctly rejects sub-threshold values with a user-facing warning.

DS-26: Prediction Logic Mismatch (High Priority)

- **Issue:** A discrepancy was found where the UI displayed a "Diabetes" result even when the model's probability score was as low as 35.09%.

- **Root Cause:** The `ML_Backend` prediction threshold was not correctly synchronized with the frontend's conditional rendering logic. The UI was triggering the "Positive" label prematurely.
- **Resolution:** Standardized the classification threshold at ≥ 0.50 across the entire pipeline. The UI now dynamically pulls the classification directly from the model's `predict` method rather than a hardcoded frontend check.
- **Verification:** Regression tested with "Borderline" edge cases. The UI result now perfectly matches the Probability Gauge output.

Defect Distribution Summary

Category	Count	Resolved	Open
High Priority Bugs	2	2	0
Medium / Low Priority Bugs	0	0	0
Total Bugs in Project	2	2	**0**

> **Verified from Jira:** JQL query `issuetype = Bug AND project = DS` returned exactly 2 results: DS-25 and DS-26, both in `Done` status. **Zero open bugs** exist in the project at the time of release sign-off.

4. Regression History

Following the resolution of DS-25 and DS-26, a full regression run was executed across all 18 sub-tasks as required by STP Section 4.4. The table below documents the status transition history for each affected test case.

TC ID	Test Case Title	Initial Run	Post-Fix Run	Linked Bug
TC-01	Validate Pregnancies field (0–20)	**PASS**	**PASS**	—
TC-02	Validate Glucose field (0–250)	**FAIL**	**PASS**	DS-25
TC-03	Validate BMI field (0–100)	**FAIL**	**PASS**	DS-25
TC-04	Validate Insulin field (0–1000)	**PASS**	**PASS**	—
TC-05	Validate Age field (0–100)	**FAIL**	**PASS**	DS-25
TC-06	Prediction with valid high-risk inputs	**PASS**	**PASS**	—
TC-07	Prediction with borderline edge cases	**FAIL**	**PASS**	DS-26

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TC ID	Test Case Title	Initial Run	Post-Fix Run	Linked Bug
TC-08	Prediction probability output sensitivity	**PASS**	**PASS**	—
TC-09	SHAP explanation generation	**PASS**	**PASS**	—
TC-10	Permutation importance display	**PASS**	**PASS**	—
TC-11	Response time for prediction (latency)	**PASS**	**PASS**	—
TC-12	UI/UX responsiveness under stress	**PASS**	**PASS**	—
TC-13	All input fields visible and accessible	**PASS**	**PASS**	—
TC-14	Model performance metrics dashboard	**PASS**	**PASS**	—
TC-15	Production deployment & CI/CD verification	**CP**	**CP**	Env. Limit
TC-16	/predict endpoint security	**PASS**	**PASS**	—
TC-17	Request/response format integrity	**PASS**	**PASS**	—
TC-18	Error handling on backend failure	**PASS**	**PASS**	—
Summary		**14P / 4F / 1CP**	**17P / 1CP**	

Legend: P = Pass, F = Fail (initial run only), CP = Conditional Pass, Env. Limit = Environment Limitation

5. Residual Risk

5.1 TC-15 — Production Deployment: Conditional PASS

Risk Classification: Environment-Level Limitation — *Not a Code Defect*

Affected Test Case: TC-15 (DS-20: 17.3 Production Deployment & CI/CD Pipeline Verification)

Risk Level: Low

Description: The DiabAI application was successfully deployed to the Railway.app cloud environment and verified as fully operational at `https://web-production-efc90.up.railway.app/`. All CI/CD pipeline runs on the `main` branch recorded a **100% build success rate**, as confirmed by GitHub Actions deployment logs.

The production URL is currently **OFFLINE** solely due to the expiration of Railway trial credits. This is an infrastructure billing constraint and constitutes *no defect in the application code, the CI/CD pipeline logic, or the ML model's functionality.*

Evidence of Successful Deployment:

- GitHub CI/CD pipeline logs confirm 100% build success rate (attached as evidence in Jira DS-20).
- All Railway deployment logs show clean process startup and model loading.

Mitigation: To restore production availability, a Railway paid plan or an alternative hosting provider (Render, Heroku, AWS) may be provisioned. No code changes are required. The `requirements.txt` and `Dockerfile` configurations are deployment-ready.

> **Audit Ruling:** This item is documented as a **Conditional PASS (COND. PASS or CP)** and does not block the `v1.0-Quality-Audit` release. The quality of the underlying application code and prediction logic has been fully validated in the local development environment and confirmed live in production prior to credit expiry.

6. Audit Sign-Off

Audit Sign-Off — v1.0-Quality-Audit

This Audit Release Summary documents the successful completion of the **DiabAI SQA Audit** for release version `v1.0-Quality-Audit`.

All **18 sub-tasks** and **5 User Stories** (DS-2, DS-8, DS-14, DS-17, DS-21) are marked **DONE** in Jira. High-priority bugs **DS-25** and **DS-26** were identified, prioritised, documented with screenshot attachments (`bug.png`, `bug 2.png`), resolved on **28 March 2026**, and confirmed via `regression testing`. The Jira release `v1.0-Quality-Audit` has been formally tagged and set to **RELEASED** status effective **29 March 2026**.

Zero open defects exist in the project. The single Conditional Pass recorded for TC-15 is an environment-level billing constraint and does not affect the integrity of the application code or ML pipeline.

The **DiabAI Diabetes Prediction System** is hereby declared **APPROVED FOR RELEASE**.

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Role:	SQA Engineer
Date:	29 March 2026
Project:	DiabAI_SQA (Jira Epic DS-1)